

Lecture 5 Solving linear first order differential equations



Lecture 5
Solving lin...

Engineering Mathematics III

ENGE702

Lecture 5: Solving linear first order differential equations

Linear first order ODEs

A first order ODE is said to be **linear** if it can be written in the form

$$y' + p(x)y = r(x).$$

For many differential equations of this type, we can solve using a technique involving what is called an **integrating factor**.

For example, the following first order ODEs are linear:

- ▶ $y' + \sin(x)y = 2x$
- ▶ $y' + 2xy - \cos(x) = 3$
- ▶ $e^x y' - 2 \ln(x)y + 5x = 2x^3$

Linear first order ODEs

Before we start to apply the integrating factor technique, it is best to put it in standard form, ie $y' + p(x)y = r(x)$. If there is a term in front of y' , we can simply divide the whole equation by that term.

Now we need to find what is called the integrating factor, defined as

$$\mu(x) = e^{\int p(x) dx}$$

This integrating factor is designed specifically such that when you multiply it through the original differential equation it becomes:

$$(\mu(x)y)' = \mu(x)r(x).$$

Note you can set the arbitrary constant in the result for $\int p(x) dx$ to be zero, since a non-zero value simply results in multiplying the whole equation by a constant which may be cancelled.

We can then integrate both sides and solve for y to get:

$$y = \frac{1}{\mu(x)} \left(\int \mu(x)r(x) dx \right)$$

Linear first order ODEs

We have then solved the differential equation - as long as we can calculate $\int \mu(x)f(x) dx$. This is not always so easy, but for many differential equations it will be possible.

Example: Solve the differential equation

$$y' + 2y = e^{-4x+7}$$

We have: $p(x) = 2$, $r(x) = e^{-4x+7}$, $\mu(x) = e^{2x}$.

So,

$$y = e^{-2x} \int (e^{2x} e^{-4x+7} dx)$$

$$y = e^{-2x} \int e^{-2x+7} dx$$

Not every 1st order ODE
is separable:

e.g. $y' = y+x$ is not separable.

this is a linear first order ODE

A general form of linear first order ODE

is $y' + p(x)y = r(x)$
 (Linear combination of dependent variables)

Steps to solve a first order linear ODE

$$y' + p(x)y = r(x)$$

Step 1: Find the integrating factor

$$\mu(x) = e^{\int p(x) dx}$$

Step 2: Multiply the integrating factor to both sides of the ODE to get

$$(\mu(x) \cdot y)' = \mu(x)r(x)$$

Step 3: Integrating both sides and solve for y as

$$\mu(x) \cdot y = \int \mu(x)r(x) dx$$

$$y = \frac{1}{\mu(x)} \left(\int \mu(x)r(x) dx \right)$$

Example: $y' + 2y = e^{-4x+7}$

(Linear combination of y and y')
 so it is a first order linear ODE.

So,

$$y = e^{-2x} \int (e^{2x} e^{-4x+7} dx)$$

$$y = e^{-2x} \int e^{-2x+7} dx$$

$$y = -\frac{1}{2} e^{-2x} e^{-2x+7} + c e^{-2x}$$

$$y = -\frac{1}{2} e^{-4x+7} + c e^{-2x}$$

Linear first order ODEs

Example: Solve the initial value problem

$$x^2 y' + 2xy = 3x + 1, \quad y(1) = 1$$

This is not in standard form, so we convert it to standard form by dividing through by x^2 to get:

$$y' + \frac{2}{x}y = \frac{3}{x} + \frac{1}{x^2}$$

We have: $p(x) = \frac{2}{x}$, $r(x) = \frac{3}{x} + \frac{1}{x^2}$, $\mu(x) = e^{\int \frac{2}{x} dx} = e^{2 \ln(x)} = x^2$.
So

$$y = x^{-2} \left(\int \left(x^2 \left(\frac{3}{x} + \frac{1}{x^2} \right) dx \right) \right)$$

$$y = x^{-2} \left(\int (3x + 1) dx \right)$$

$$y = x^{-2} \left(\frac{3x^2}{2} + x \right) + c x^{-2}$$

$$y = \frac{3}{2} + \frac{1}{x} + \frac{c}{x^2}$$

Linear first order ODEs

Now we apply the initial condition: $y(1) = 1$, so:

$$1 = \frac{3}{2} + \frac{1}{1} + \frac{c}{1^2}$$

$$1 = \frac{3}{2} + 1 + c$$

$$c = -\frac{3}{2}$$

Therefore, the particular solution is

$$y = \frac{3}{2} + \frac{1}{x} - \frac{3}{2x^2}$$

Linear first order ODEs

Example: An electrical circuit has an EMF of $E = 48V$ (constant), a resistor of $R = 11\Omega$ and an inductor of $L = 0.1H$. The current is initially 0. Find the current in the circuit as a function of time.

To solve this we need some laws of physics. The first is Ohm's Law: a current in the circuit causes a voltage drop across the resistor of IR , so $V_1 = IR$.

The second is Kirchhoff's Law: a current in the circuit causes a voltage drop across the inductor of $L I' = L \frac{dI}{dt}$, so $V_2 = L I'$.

Finally, the sum of these two voltage drops gives the EMF, so $E = V_1 + V_2$, or

$$L I' + R I = E.$$

Linear first order ODEs

We will solve this differential equation. We will keep the numbers given as letters for now - this won't make it much harder to do, and may come in handy later if we want to change things.

First we need to put the equation in standard form:

$$I' + \frac{R}{L} I = \frac{E}{L}.$$

Here we have: $p(t) = \frac{R}{L}$, $r(t) = \frac{E}{L}$, and $\mu(t) = e^{\int \frac{R}{L} dt} = e^{\frac{R}{L} t}$.
So,

$$I = e^{-\frac{R}{L} t} \left(\int e^{\frac{R}{L} t} \frac{E}{L} dt \right) + c e^{-\frac{R}{L} t}$$

$$I = e^{-\frac{R}{L} t} \left(\frac{E L}{-L} e^{\frac{R}{L} t} \right) + c e^{-\frac{R}{L} t}$$

linear combination of y and y'
so it is a first order linear ODE.

step 1: integrating factor

$$\mu(x) = e^{\int p(x) dx} = e^{\int 2 dx} = e^{2x}$$

step 2: multiply integrating factor to both sides of ODE to get

$$(e^{2x} y)' = e^{-4x+7} \cdot e^{2x} = e^{-2x+7}$$

step 3: Integrate both sides

$$e^{2x} y = \int e^{-2x+7} dx = -\frac{1}{2} e^{-2x+7} + C$$

$$\Rightarrow y = -\frac{1}{2} e^{-4x+7} + C e^{-2x}$$

Example $x^2 y' + 2xy = 3x + 1$, $y(1) = 1$

divided by $\Rightarrow y' + \frac{2}{x}y = \frac{3}{x} + \frac{1}{x^2}$

integrating factor: $\mu(x) = e^{\int p(x) dx} = e^{\int \frac{2}{x} dx} = e^{2 \ln x} = e^{\ln x^2} = x^2$

multiply the integrating factor to both sides:

$$(x^2 y)' = \left(\frac{3}{x} + \frac{1}{x^2} \right) x^2 = 3x + 1$$

integrate both sides $x^2 y = \int (3x + 1) dx = \frac{3}{2} x^2 + x + C$

$$\Rightarrow y = \frac{3}{2} + \frac{1}{x} + \frac{C}{x^2}$$

$$y(1) = \frac{3}{2} + \frac{1}{1} + \frac{C}{1} = 1 \Rightarrow C = -\frac{3}{2}$$

so the particular solution is

$$y = \frac{3}{2} + \frac{1}{x} - \frac{3}{2x^2}$$

Example: for an electrical circuit, we can use Kirchhoff's law to get

$$L I' + R I = E, \quad \boxed{L, R, E \text{ are constants}}$$

first order linear ODE.

$$I' + \frac{R}{L} I = \frac{E}{L}$$

Integrating factor: $\mu(t) = e^{\int \frac{R}{L} dt} = e^{\frac{R}{L} t}$

multiply both sides $(e^{\frac{R}{L} t} \cdot I)' = \frac{E}{L} e^{\frac{R}{L} t}$

integrate both sides $e^{\frac{R}{L} t} \cdot I = \int \frac{E}{L} e^{\frac{R}{L} t} dt$

So,

$$I = e^{-\frac{R}{L}t} \left(\int e^{\frac{R}{L}t} \frac{E}{L} dt \right) + c e^{-\frac{R}{L}t}$$

$$I = e^{-\frac{R}{L}t} \left(\frac{E}{L} \frac{L}{R} e^{\frac{R}{L}t} \right) + c e^{-\frac{R}{L}t}$$

$$I = \frac{E}{R} + c e^{-\frac{R}{L}t}$$

Linear first order ODEs

Inputting our values for E , R , and L , we have

$$I = \frac{48}{11} + c e^{-110t}$$

Now we can put in the initial condition $I(0) = 0$, and get

$$0 = \frac{48}{11} + c e^0$$

$$c = -\frac{48}{11}$$

so our current over time becomes

$$I = \frac{48}{11} (1 - e^{-110t})$$

integrate both sides $\int e^{\frac{R}{L}t} \cdot I = \int \frac{E}{L} e^{\frac{R}{L}t} dt$

$$\Rightarrow e^{\frac{R}{L}t} \cdot I = \frac{E}{L} \cdot \frac{L}{R} e^{\frac{R}{L}t} + c$$

$$\Rightarrow e^{\frac{R}{L}t} \cdot I = \frac{E}{R} e^{\frac{R}{L}t} + c$$

$$\Rightarrow I(t) = \frac{E}{R} + c \cdot e^{-\frac{R}{L}t}$$

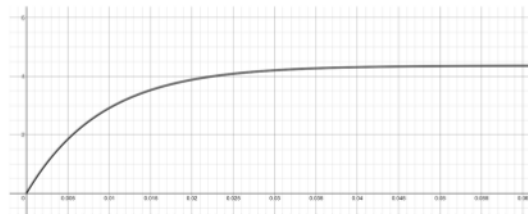
$$E = 48, R = 11, L = 0.1$$

$$\text{so } I(t) = \frac{48}{11} + c \cdot e^{-110t}$$

$$I(0) = \frac{48}{11} + c e^0 = 0 \Rightarrow c = -\frac{48}{11}$$

$$\text{so } I(t) = \frac{48}{11} (1 - e^{-110t})$$

Linear first order ODEs



Exercises

► Solve the following ODEs:

► $y' + xy = y$

► $y' + \frac{3x}{2}y = 2x$

► A bath is run to a temperature of 40° . The temperature in the bathroom is 18° . After 15 minutes the bath-water is 36° . How long before the water reaches 33° ?

Text Book References

Chapter 1.5